Computer Hardware

1. Hardware

Definition:
Hardware is the physical part of a computer that can be seen and touched. It works with software to perform tasks.

Examples:

- Motherboard
- CPU
- RAM
- Hard Disk/SSD
- Monitor
- Keyboard
- Mouse
- Power Supply

Motherboard Ports and Components

2. PS/1 Port

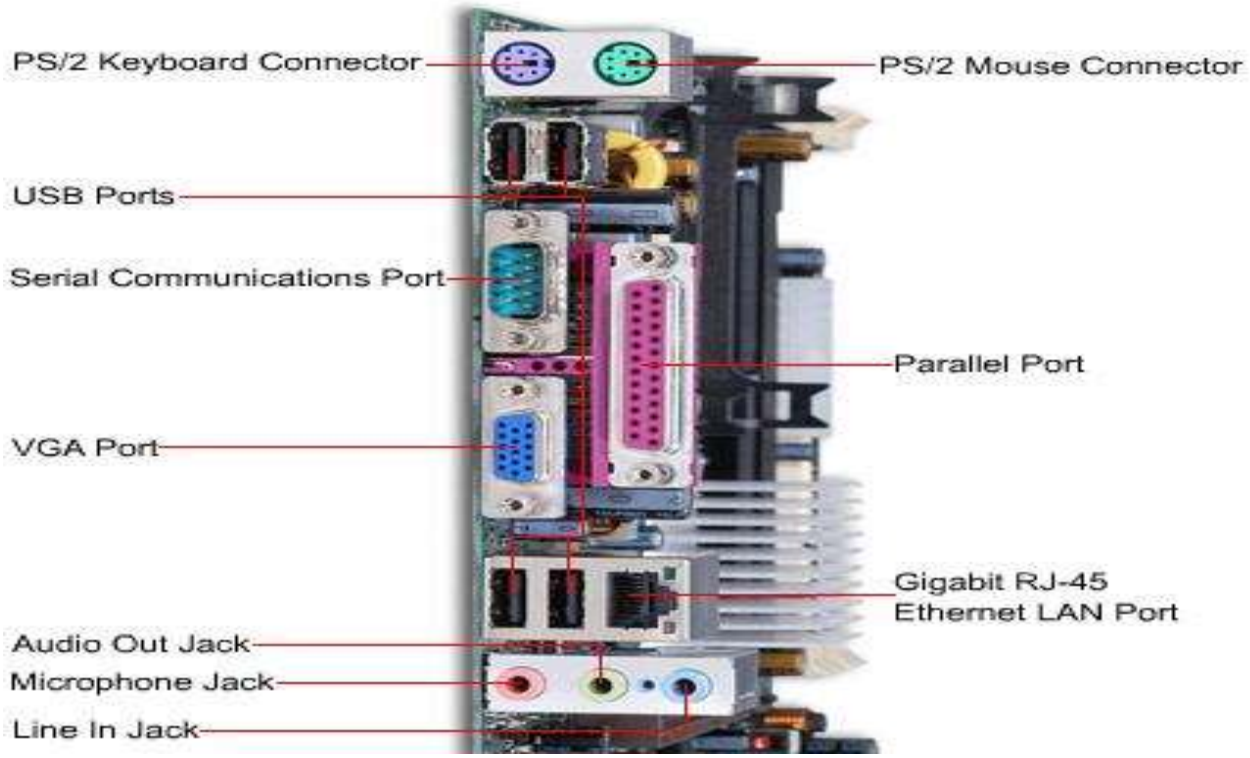
Definition:
PS/1 was an IBM computer standard. It is not a common port like PS/2. Sometimes people mistakenly write PS/1 instead of PS/2.

Uses:

- Used in old IBM PS/1 computers.
- Rarely used today.

---

3. PS/2 Port


PS/2 Keyboard Connector
PS/2 Mouse Connector
USB Ports
Serial Communications Port
Parallel Port
VGA Port
Gigabit RJ-45
Ethernet LAN Port
Audio Out Jack
Microphone Jack
Line In Jack





3. PS/2 Port:

Definition:
PS/2 is a 6-pin mini-DIN connector used to connect a keyboard and mouse.

Color

- Purple → Keyboard
- Green → Mouse

Advantages

- Stable connection
- Doesn't require USB
- Works during BIOS setup

Disadvantages

- Old technology
- Hot-plugging is not recommended

---

4. Serial Port (COM Port)

Definition:
A Serial Port transfers data one bit at a time.

Connector

- DB-9 (9 pins)
- DB-25 (25 pins)

Uses

- Old Mouse
- Modem
- Barcode Scanner
- Industrial Machines

Advantages

- Reliable
- Long-distance communication

Disadvantages

- Slow compared to USB

---

5. DVI (Digital Visual Interface)

(Often written as DVT by mistake.)

Definition
DVI is a video display interface.

Types

- DVI-D
- DVI-I
- DVI-A

Uses

- LCD Monitor
- Computer Monitor

Advantages

- Better quality than VGA

---

6. VGA (Video Graphics Array)

Definition
VGA is an analog display connector.

Pins

- 15 Pins

Color

- Blue connector

Uses

- Monitor
- Projector

Advantages

- Supported by older devices

Disadvantages

- Lower image quality than HDMI or DisplayPort

---

7. USB (Universal Serial Bus)

Definition
USB connects peripherals to a computer.

Versions

- USB 1.1
- USB 2.0
- USB 3.0
- USB 3.1
- USB 3.2
- USB4

Uses

- Keyboard
- Mouse
- Printer
- Pen Drive
- External HDD
- Webcam

Advantages

- Plug and Play
- Fast data transfer
- Power supply through cable

---

8. LAN Port (RJ-45)

Definition
LAN Port connects a computer to a network.

Connector

- RJ-45

Uses

- Internet
- Office Network
- File Sharing

Speed

- 100 Mbps
- 1 Gbps
- 2.5 Gbps
- 10 Gbps

---

9. Sound Port (Audio Port)

Definition
Audio ports connect speakers, microphones, and headphones.

Colors

- Green → Speaker
- Pink → Microphone
- Blue → Line In

Internal Components




华夏浩风



10. Power Supply (SMPS)

Definition
SMPS (Switch Mode Power Supply) converts AC electricity into DC voltage for computer components.

Output Voltages

- +12V
- +5V
- +3.3V

Functions

- Supplies power to motherboard
- CPU
- Hard Disk

- Graphics Card
- Fans

---

11. Heat Sink

Definition
A Heat Sink removes heat from electronic components.

Made of

- Aluminum
- Copper

Uses

- CPU
- GPU
- Chipset

Purpose

- Prevent overheating
- Improve performance
- Increase hardware life

---

12. CPU (Dual Core Processor)

Definition
CPU is the brain of the computer.

Dual Core
A Dual Core processor contains **2 processing cores**.

Advantages

- Better multitasking
- Faster than single-core processors
- Lower power consumption

---

13. RAM (Random Access Memory)

Definition
RAM is temporary memory used while the computer is running.

Types

- DDR
- DDR2
- DDR3
- DDR4
- DDR5

Characteristics

- Volatile memory
- Very fast
- Stores running programs

---

14. IDE (Integrated Drive Electronics) / PATA

Definition
IDE (also called PATA) is an old interface for connecting storage devices.

Cable

- 40-pin ribbon cable

Uses

- Hard Disk
- CD/DVD Drive

Disadvantages

- Slow
- Large cable

---

15. SATA (Serial ATA)

Definition
SATA replaced IDE/PATA.

Advantages

- Faster
- Smaller cable
- Better airflow
- Easy installation

Uses

- HDD
- SSD
- DVD Drive

Common Speeds

- SATA I → 1.5 Gbps
- SATA II → 3 Gbps
- SATA III → 6 Gbps

---

16. PCI (Peripheral Component Interconnect)

Definition
PCI is an expansion slot used to add hardware.

Cards Installed

- LAN Card
- Sound Card
- USB Card
- TV Tuner Card

---

17. AGP (Accelerated Graphics Port)

Definition
AGP was designed only for graphics cards.

Advantages

- Faster than PCI
- Dedicated graphics slot

Status
Now obsolete.

---

18. PCI Express (PCIe)

Definition
PCIe replaced AGP and PCI.

Uses

- Graphics Card
- NVMe SSD
- Wi-Fi Card
- Capture Card
- Network Card

Advantages

- Very high speed
- Multiple lane configurations (×1, ×4, ×8, ×16)

---

19. External Graphics Card (GPU)

Definition
A Graphics Processing Unit (GPU) processes images, videos, and 3D graphics.

Uses

- Gaming
- Video Editing
- CAD
- AI
- Machine Learning

Manufacturers

- NVIDIA
- AMD
- Intel

---

20. BIOS Battery (CMOS Battery)

Definition
The BIOS battery (CMOS battery) powers the motherboard's CMOS memory and real-time clock when the computer is turned off.

Battery Type

- CR2032 (3V Lithium Coin Cell)

Functions

- Saves BIOS settings
- Maintains date and time

Symptoms of Failure

- Wrong date/time
- BIOS settings reset
- CMOS checksum error

Additional Important Hardware Topics

21. Motherboard

The motherboard is the **main circuit board** that connects all computer components, including the CPU, RAM, storage devices, expansion cards, and ports.

22. Storage Devices

- **HDD (Hard Disk Drive):** Uses spinning magnetic disks; offers high capacity at a lower cost.
- **SSD (Solid State Drive):** Uses flash memory; much faster, quieter, and more durable than HDDs.
- **NVMe SSD:** A high-speed SSD that connects through PCIe for significantly faster performance.

23. Cooling Fan

A cooling fan improves airflow and helps remove heat from components such as the CPU, GPU, and power supply.

24. Optical Drive

An optical drive reads and writes CDs, DVDs, or Blu-ray discs. It is less common in modern computers.

25. CMOS

CMOS (Complementary Metal-Oxide Semiconductor) is a small memory chip that stores BIOS/UEFI settings such as boot order and hardware configuration. It is powered by the CMOS battery.